

(a) Training: a flow-matching prior on the GEOMETRY bridge

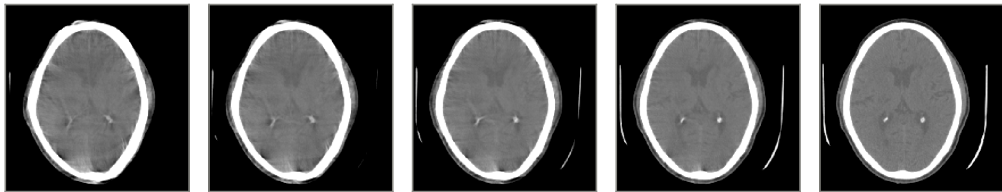
$t = 0$

$t = 0.25$

$t = 0.5$

$t = 0.75$

$t = 1$



$x_t = \text{FDK}(A(x; P_{\text{nom}} T((1-t)\theta)), P_{\text{nom}})$ $v_\phi(x_t, t) \approx dx_t/dt$ — the target is ANALYTIC (the FORWARD projector's exact geometry derivative)

Every point is an FDK reconstruction carrying exactly $(1-t)$ of the true motion, so the path ends on the motion-free static FDK BY CONSTRUCTION.

(b) Inference: blind rigid-motion correction — one Euler step of $N = 50$ (θ and x bootstrap each other)

cold start: $x \leftarrow \text{FDK}(y, P_{\text{nom}})$, $\hat{\theta} \leftarrow 0$

1. PREDICT — the prior moves

$$x_{\text{pred}} = x + \Delta t v_\phi(x, t)$$

frozen UNet, evaluated patch-wise
and blended over 2 tilings

2. ESTIMATE the motion

$$\theta = \arg \min_{\theta} \|A_{\theta} x_{\text{pred}} - y\|^2$$

6-DoF per view, hash-encoding MLP,
200 iters/step, 24 views/iter, warm start

3. CORRECT — PnP data step

$$z \leftarrow \arg \min_z \|A_{\hat{\theta}} z - y\|^2 \quad (5 \text{ CG its})$$

$$z \leftarrow z + \kappa (\text{TV}(z) - z)$$

$$x \leftarrow z, \quad t \leftarrow t + \Delta t$$

Gauss–Seidel: the motion is always fitted on the image the prior just improved